



Inertial effects of a shock accelerated particle on the time dependent drag coefficient

Criston Hyett¹, Iman Borazjani^{*2}, and Daniel Livescu^{†1}

¹*Computational Physics and Methods, CCS-2, Los Alamos National Laboratory, Los Alamos, NM, USA*

²*J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University, College Station, TX, USA*

Despite a long history of theory and experimentation of shock-particle interaction, elucidating the time-dependent drag coefficients for single and multi-particle systems accelerated by shocked flow remains an open challenge. Numerical simulations are uniquely positioned as a relatively inexpensive alternative to experimentation to explore the parameter space of shock accelerated particle flows; supplying critically needed data to time-dependent drag models. We utilize the flexibility and efficiency of a sharp-interface immersed boundary method (SI-IBM) to extract a high-fidelity approximation to the time-dependent drag coefficient $C_d(t)$ of a spherical particle accelerated by a shock. Critically, the SI-IBM allows the particle to accelerate and interact with its wake. We examine how variations of Reynolds number and density ratio affect $C_d(t)$ for this Lagrangian particle, and compare the results to previous Eulerian simulations. Finally, we utilize system identification techniques to construct a parameterization of the correction originating from particle motion.

Nomenclature

d_p	particle diameter
Re	Reynolds number
C_d	drag coefficient
ρ	density
μ	dynamic viscosity
p	pressure
m	mass
u	fluid velocity
v	particle velocity
c	speed of sound
$(\cdot)_p$	particle value
$(\cdot)_L$	post-shock value
$(\cdot)_R$	pre-shock value

I. Introduction

Particle-laden, high speed flows have long been studied: beginning with Newton who postulated regarding subsonic flow over observable spheres; becoming the subject of study of ballistics of subsonic cannonshot; eventually refining toward supersonic gas flows and microscopic particles, having application in rocket engines and supernova[1]. Despite this long history of exploration, physical descriptions of the transient behavior of a shock-accelerated particle are still lacking. Experiments continue to add to our body of knowledge, but accurately resolving short-time dynamics of small (micro) particles is costly and difficult[2]. We point to simulation as a relatively inexpensive alternative, and propose the use of a sharp-interface immersed boundary method to simulate the shock-particle interaction.

For simplicity, our study is confined to spherical particles in high-speed (compressible) flows. In particular, we investigate plane-wave shocks impinging upon a single spherical particle initially at rest. These flows can be

^{*}Professor, AIAA Associate Fellow

[†]Scientist, AIAA Associate Fellow

characterized by a small set of nondimensional parameters - described fully in SecII. Our consideration is restricted to continuum, non-turbulent, compressible flows, with sub-critical particle Mach number. Much of the previous work augmented experimental data using simulation [3–6], where the particle was fixed, and/or ignored transient effects [6]. We maintain that simulating the truly Lagrangian nature of the interaction will allow extension of the work beyond highly symmetric particles, and through larger ranges of the parameter space that may contain longer relaxation times. One previous study [7] did perform a Lagrangian simulation for a single set of parameters, and their results motivate our adiabatic assumption of the particle.

We begin the manuscript with a basic description of the physics governing shock-particle interaction, and a brief overview of the highly parallelized, sharp-interface immersed boundary method used to resolve the coupled system of partial differential equations. The results of our simulations are compared against the phenomenological Parmar model [4] (previously shown to agree well with experimental data[3]). Simulations across a range of defining parameters are performed, and the time-dependent drag coefficient $C_D(t)$ for the shock-accelerated particle is collected. Toward the dual goal of disseminating the data effectively and providing phenomenological insight, we construct a targeted, scientific machine learning extension of the Parmar model. Taken together, the high-fidelity simulation data and extended phenomenological model provide additional insights into the unsteady forces on the particle, and dependence of transient dynamics on parameters.

II. Methods

Frictional gas dynamics are governed by compressible Navier-Stokes, yielding six equations: conservation of mass, three for conservation of momentum, and conservation of energy:

$$\partial_t \rho + \nabla(\rho \mathbf{u}) = 0 \quad (1)$$

$$\partial_t(\rho \mathbf{u}) + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla \cdot \hat{\sigma} = 0 \quad (2)$$

$$\partial_t(\rho E) + \nabla \cdot ((\rho E + p) \mathbf{u}) - \nabla \cdot \mathcal{T} = 0 \quad (3)$$

where $\mathcal{T}$ is the gradient of internal energy defined by:

$$\mathcal{T} = \gamma \mu \text{Pr}^{-1} \nabla \left(E - \frac{1}{2} \mathbf{u} \cdot \mathbf{u} \right) \quad (4)$$

and σ_{ij} the fluid stress tensor:

$$\sigma_{ij} = -p\delta_{ij} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right). \quad (5)$$

These equations are supplemented by the ideal gas law, relating pressure, density and temperature:

$$pV = nRT \quad (6)$$

An immersed solid (i.e., without deformation) in the fluid undergoes motion dictated by Newton's second law:

$$\partial_t \mathbf{x}_p = \mathbf{v} \quad (7)$$

$$m_p \partial_t v_i = \oint_s \sigma_{ij} n_j dS \quad (8)$$

where $\mathbf{x}_p$, $\mathbf{v}$ are the particle position and velocity. Equations (1-3, 7) are coupled to eq(8) by evaluation of the fluid stress tensor at the particle surface. The coupling is two-way [8, 9] as the solid impedes the flow of the fluid, inducing changes in the pressure and flow fields.

For shocked flow, the Rankine–Hugoniot relations relate pressure, density and velocity across the shock front for a given shock Mach number, M_s :

$$\frac{p_L}{p_R} = M_s^2 + \eta (M_s^2 - 1) \quad (9)$$

$$\frac{\rho_L}{\rho_R} = \frac{1 + \frac{p_L/p_R}{\eta}}{1/\eta + p_L/p_R} \quad (10)$$

$$\frac{u_L}{c_R} = \frac{p_L/p_R - 1}{\gamma M_s} \quad (11)$$

Boundary	Condition	Equations
$(x = \pm \frac{L_x}{2}, y, z)$	“slip”, Nuemann	$u_x = 0; \partial_x(\rho, u_y, u_z, E) = 0$
$(x, y = \pm \frac{L_y}{2}, z)$	“slip”, Neumann	$u_y = 0; \partial_y(\rho, u_x, u_z, E) = 0$
$(x, y, z = 0)$	Inflow	$u_x, u_y = 0; \partial_z(\rho, u_z, E) = 0$
$(x, y, z = L_z)$	Outflow	$\partial_z(\rho, u_x, u_y, u_z, E) = 0$
particle surface	No-slip	$\mathbf{u} = \mathbf{v}$

Table 1 Boundary conditions

Where γ is the ratio of specific heat ratio, $\eta := \frac{\gamma-1}{\gamma+1}$, and it is assumed $u_R = 0$. $(\cdot)_R, (\cdot)_L$ denote pre- and post-shock values of pressure p , gas velocity u and density ρ , and speed of sound c .

The simulation of eqs(1-8) is characterized by three nondimensional numbers: the particle Reynolds (Re_p) and Knudsen (Kn) numbers, and the density ratio (β):

$$Re_p = \frac{\rho_L d_p (u_L - v)}{\mu}, \quad Kn = \frac{\lambda}{d_p}, \quad \beta = \frac{\rho_p}{\rho_L} \quad (12)$$

with λ the mean-free path length in the post-shock gas, and $v := \|\mathbf{v}\|$. Our simulation is predicated on Navier-Stokes and is thus only valid in the continuum regime, defined by $Kn \leq 0.01$, restricting our choice of particle diameter to $d_p \geq 4e-6$. Further, we assume turbulent effects can be neglected, justified by the relatively small Re under consideration.

The fluid-structure interaction is solved utilizing the curvilinear immersed boundary (CurvIB) method [8], advanced with 3rd order Runge-Kutta timestepping. The sharp interface at the immersed boundaries are maintained by interpolating variables along the normal direction to the boundary [8]. Full description of the numerical implementation can be found in [8, 10]. From this methodology, two natural restrictions on the timestep arise: the usual Courant-Friedrich-Levy (CFL) condition, and a restriction that the immersed solid move less than a grid-spacing per timestep. In the parameter regime we are investigating - namely where the particle velocity is small compared to the post-shock gas velocity - the CFL condition is the more restrictive of the two conditions. We use a Euclidean, stretched grid discretization, handled by generalized curvilinear transformation [8], to allow for fine resolution near the sphere, and a coarse discretization near the boundaries, simultaneously increasing fidelity of the shock-particle interaction, reducing simulation cost, and dissipating any numerical reflections from the boundary conditions. For our simulation, a numerical study suggested a stretched-grid discretization ranging from $0.02d_p$ near the particle trajectory to $0.5d_p$ at the boundaries. A schematic of the simulation setup is found in Fig[1]. The boundary conditions are listed in Table[1].

The time-varying force (neglecting lift and bouyancy) on a particle in compressible flow can be decomposed as[4]:

$$\mathbf{F}(t) = \mathbf{F}_{qs}(t) + \mathbf{F}_{iu}(t) + \mathbf{F}_{vu}(t) \quad (13)$$

with the total force decomposed into quasi-steady F_{qs} , inviscid unsteady F_{iu} , and viscous unsteady F_{vu} forces. While considerable effort has described the quasi-steady force F_{qs} - see e.g. [6] - the unsteady forces have largely gone unmodeled. A line of work [4, 5, 11] approached the modeling task for the inviscid unsteady force F_{iu} , neglecting F_{vu} due to known complications and because the created model described previous simulation[3] well. In [4], the authors construct an approximation of F_{iu} via a temporal convolution of a kernel $K(\tau)$, and the time history of relative acceleration between the particle and the surrounding gas; a functional form thought to be accurate to describe F_{vu} as well. The developed Parmar model is described fully in Section VI, but they model the inviscid unsteady term as an integro-differential equation (IDE):

$$F_{iu}(t) = \int_{-\infty}^t K_{sp} \left(\frac{c_\infty}{R} (t - \chi) \right) \cdot \left(\frac{D(m_f u)}{D\chi} - \frac{d(m_f v)}{d\chi} \right) d \left(\frac{c_\infty}{R} \chi \right) + \int_V \rho_f \frac{Du}{Dt} dV \quad (14)$$

For our purposes, as we are unable to separate viscid and inviscid contributions, we choose a different decomposition of the total force:

$$\mathbf{F}(t) = \mathbf{F}_{qs}(t) + \mathbf{F}_{pg} + \mathbf{F}_{hist} \quad (15)$$

where we carve out the pressure gradient term F_{pg} from the inviscid unsteady term, to obtain $F_{hist}(t) := F_{iu} + F_{vu} - F_{pg}$, which we consider an aggregation of forces requiring convolution of the history of relative acceleration with a kernel.

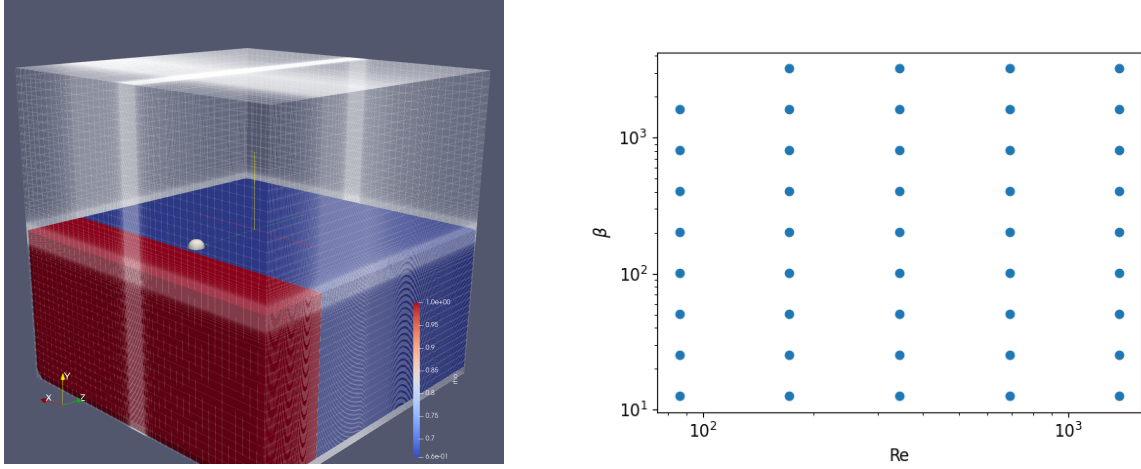


Fig. 1 (Left) Schematic of simulation setup. Example refinement is shown by wireframe shading, with high resolution near the shock-particle interaction, and along the particle trajectory. We use a coarse grid near the boundaries to dissipate numerical reflections. A shock is initialized upwind of the spherical particle, illustrated by a cutout of the piecewise constant density profile. (Right) The parameter space explored to date. An interesting extension of this is along the shock Mach axis.

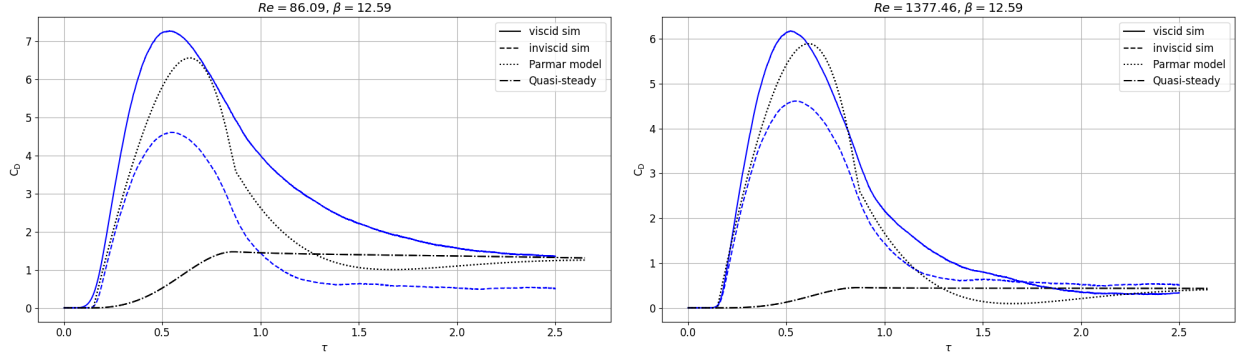


Fig. 2 Comparison of time-varying drag coefficient in a low β regime. Results are shown for viscid and inviscid simulations, as well as for Parmar and the quasi-steady drag coefficients.

III. Results

We begin this section with a discussion of the simulation results and exhibit differences with previous simulations and phenomenological models. Then, a scientific machine learning methodology is pursued to learn a parameterized kernel $K(\tau) \rightarrow K_\theta(\tau)$ for F_{hist} to extend the Parmar model to more accurately describe our simulation data with a moving particle.

To more clearly show differences between hyperparameters explored (recall Fig 1), instead of plotting (dimensional) force directly, we plot the nondimensional, time-dependent drag coefficient $C_D(t)$, which can be found from the total force by Newton's second law as:

$$C_D(t) = \frac{4\rho_p d_p}{3\rho_g} \frac{a_p(t)}{(u - v_p(t))^2}. \quad (16)$$

with $a_p(t) = F(t)/m_p$. Results from the extreme points of our parameter sweep can be found in Figs 2,3. Included in these plots are results from an inviscid simulation, the Parmar reduced-order model, and the viscid simulation of the full NSEs. The Parmar model is a reasonable stand-in for previous simulations with fixed particle. We will not discuss the inviscid results here, simply note that future work may utilize them to more directly target models of $F_{iu}(t)$ vs $F_{vu}(t)$, instead of aggregating their effects into $F_{hist}(t)$ as we will do shortly.

A few important features of these are worth highlighting. First, after the peak force occurs at half an acoustic

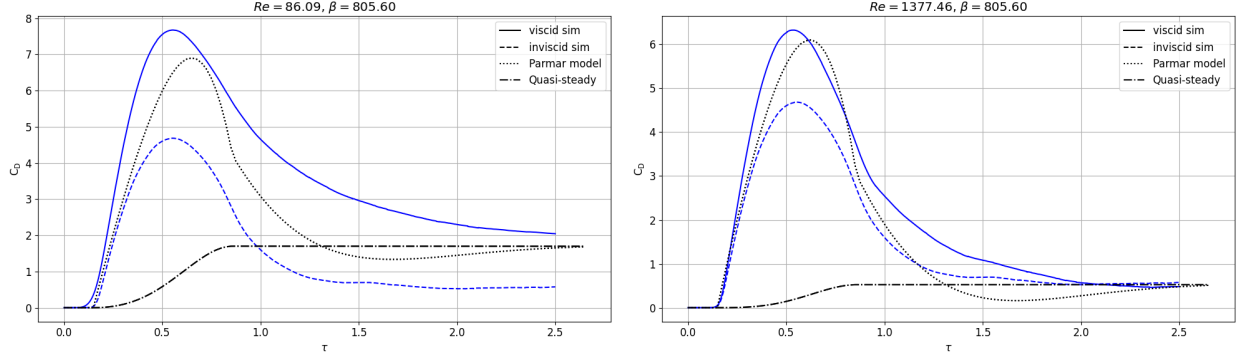


Fig. 3 Comparison of time-varying drag coefficient in a typical (for gas/solid interactions) β regime. Results are shown for viscid and inviscid simulations, as well as for Parmar and the quasi-steady drag coefficients.

timescale 0.5τ , our simulation predicts a near monotonic decrease in the unsteady forces as the particle relaxes towards the quasi-steady drag. This is in contrast to [3], which find the shock focuses behind the particle, creating a short-lived high-pressure region behind the particle, resulting in a local minimum (displayed in the Parmar model in Fig 3 at approximately 1.6τ). For very high Re , we see this effect in our simulation, albeit significantly less emphatically, around 2.25τ . We postulate that the freely moving particle likely inhibits this highly symmetric focus at the rear of the particle, and instead a complicated, coupled interaction between particle, wake and shock front occurs at these small times. Next, note the difference between the simulation and Parmar model is almost always positive. This monotonic difference results in an aggregate additional impulse to the particle, perhaps improving those models that substitute resolution of these short time dynamics with an initial velocity found by integrating the transient acceleration. Finally, we note that the parameter β seems to be considerably less important than Re , and that our results and the Parmar model come into closer alignment at higher Re .

Towards the goals of more readily sharing our results, improving transient modeling, and enriching the phenomenology of the field, we apply targeted, scientific machine learning to parameterize history kernels $K_\theta(\tau)$ to model our aggregated history force:

$$\hat{F}_{hist}(t) = \int_{-\infty}^t K_\theta \left(\frac{c_\infty}{R} (t - \chi) \right) \cdot \left(\frac{D(m_f u)}{D\chi} - \frac{d(m_f v)}{d\chi} \right) d \left(\frac{c_\infty}{R} \chi \right) \quad (17)$$

Note that rearranging Eq. 15, we can calculate the observed history force

$$F_{hist}(t) = F_{\text{total, simulated}}(t) - (F_{qs}(t) + F_{pg}(t)) \quad (18)$$

Then, for each point in our parameter sweep (Re, β) , we obtain $F_{hist}(t)$, see Fig. 4 for a representation. We then use a radial basis expansion to approximate $K_\theta(\tau)$:

$$K_\theta(\tau) = \sum_{i=1}^N \alpha_i \exp \left[-\epsilon_i^2 \left(\tau - c_i^2 \right) \right] \quad (19)$$

Due to the intricate coupling of terms in the Parmar IDE model, in order to accurately and efficiently train K_θ , one needs to calculate sensitivities of a loss:

$$\mathcal{L}(\theta) = \|F_{hist}(t) - \hat{F}_{hist}(t)\|_2^2 \quad (20)$$

with respect to the parameters θ through the solution of the IDE in Eq 14. We find the adjoint method, together with automatic differentiation to be by far the most efficient method to perform this optimization; see [12] for a review of methods. The efficiency of our implementation leverages Julia's SciML ecosystem, namely [13] and [14].

The approximation $\hat{F}_{hist}(t)$ found via convolution with the learned parameterized kernel after training can be seen in Fig. 6. The fit is much improved over the baseline Parmar model, importantly, the total impulse applied to the particle via the learned history force does replicate that from the simulation. However, modeling $\hat{F}_{hist}$ solely via parameterization of $K_\theta(\tau)$ does not allow enough flexibility to account for all the variation in F_{hist} derived from simulation. In particular,

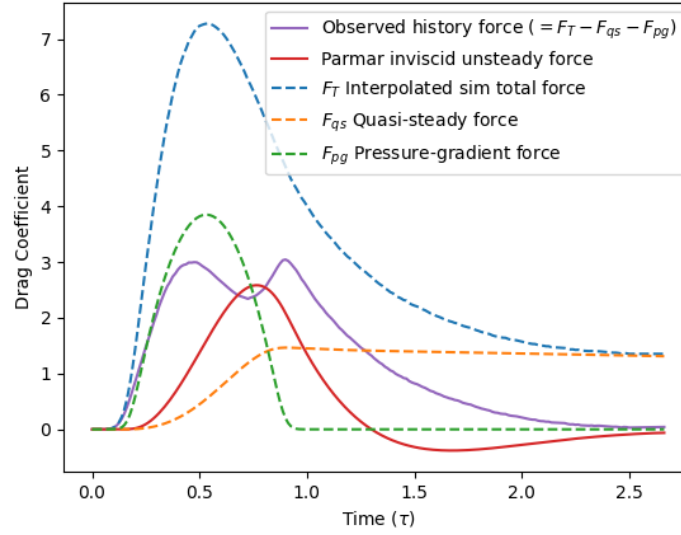


Fig. 4 Comparison of the observed history and Parmar unsteady forces (solid). Dashed are the total simulated, the idealized pressure gradient, and quasi steady forces.

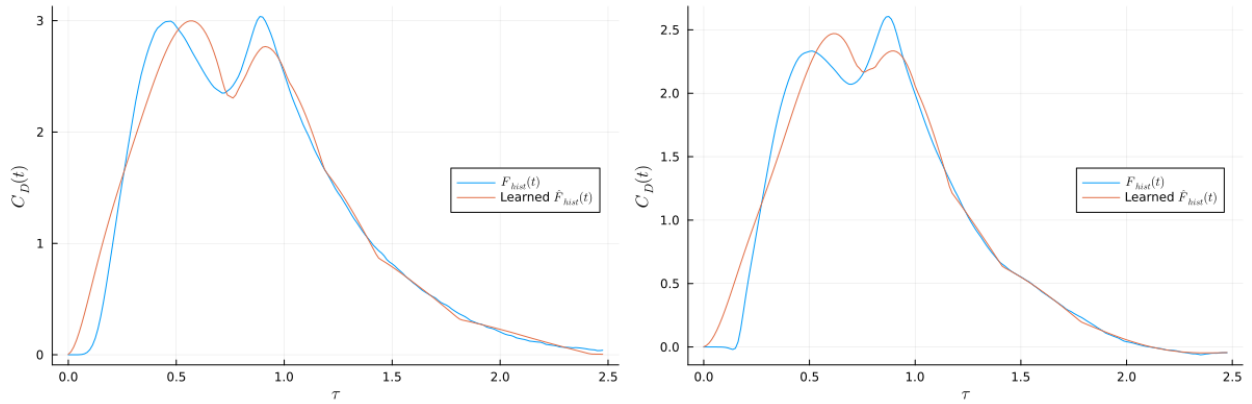


Fig. 5 We show the fit of the learned history forces to those found in the simulation for two extremes of our parameter exploration: (Left) $Re = 86, \beta = 805$ (Right) $Re = 1377, \beta \approx 5e4$

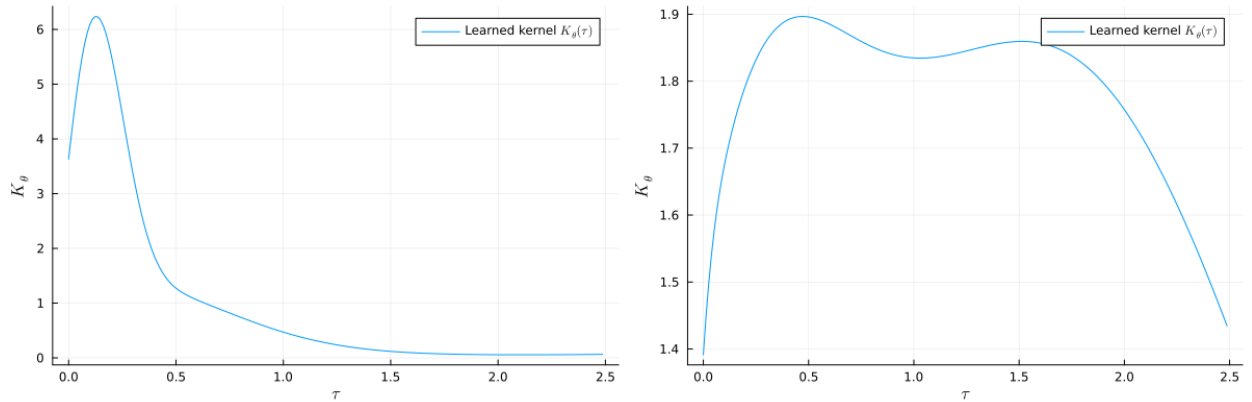


Fig. 6 The optimized history kernels $K_\theta(\tau)$, producing the resulting history force for the two scenarios in Fig 5.

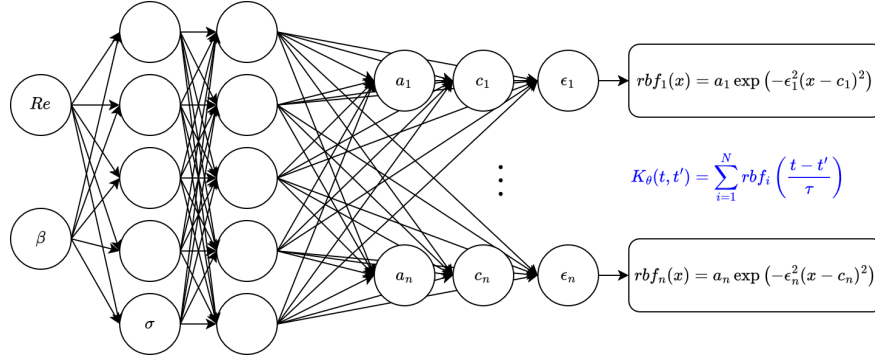


Fig. 7 A diagram of the feed forward neural network proposed to interpolate between the parameterizations of K_θ .

despite additional model flexibility, the learned kernel fails to match the amplitude of the observed double peaked history force.

All data, code, and trained parameterizations are available upon request.

IV. Conclusion

In this manuscript, we extended the literature by providing novel, time-varying estimates of drag coefficient for a Lagrangian shock-accelerated particle across a range of defining parameters, calculated via a sharp-interface immersed boundary method. Our results show qualitative agreement with previous results, but important differences point to the need for resolution of contended points. Of particular note is the lack of local minimum in $C_D(t)$, found in previous work to be caused by shock focusing at the rear of the particle. Further work must be done to disambiguate numerical (shock-capturing) differences between our and previous simulations and the effects of Lagrangian vs fixed particles used in each respectively. Additionally, direct comparisons to previous simulation and experimental data will more accurately contextualize this contribution.

Through differentiable programming and efficient calculation of gradients, we provided an extension of the Parmar model via parameterization of the history kernel $K_\theta(\tau)$. We found this term could explain much, but not all of, the difference between our results and previous models. The curious double peak in the observed history force $F_{hist}(t)$ may be attributable to the differences between an idealized pressure gradient force (used here), and the numerical pressure gradient, which will have an extended effect from the development and interaction of the shock front with the viscous boundary layer around the particle. As an example of the differing timescales, observe the difference in first derivative of force between viscous and inviscid simulation in Fig. 2 at low Re . Future work to extract the pressure gradient from the numerical simulation may reduce or eliminate the double peak of $F_{hist}(t)$, allowing for a simpler and interpretable parameterized model for $K_\theta(\tau)$.

Finally, to promote regularity of the kernels in transitions between individual parameterizations, (e.g. (Re_1, β_1) , (Re_2, β_2)), ongoing work wraps the individual parameterization in a simple feed-forward neural network, a schematic of which is found in Fig. 7. The continuity of the output from the feed-forward network encourages similarity between kernels where possible.

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VI. Appendix

A. Parmar Inviscid Unsteady Model

For completeness, this section describes the joint work of [11], and [4] to create what we refer to as the “Parmar” model.

The authors first reduce the full set of forces acting on the particle

$$F(t) = F_{qs} + F_{iu} + F_{vu} + F_l + F_{bg} \quad (21)$$

where F_{qs} is the quasi-steady, F_{iu} is the inviscid unsteady, F_{vu} is the viscid unsteady, F_l is lift, and F_{bg} is the buoyancy/gravity forces. The authors use symmetry to discard lift, and disregard buoyancy and viscous unsteady forces. Thus,

$$F(t) = F_{qs} + F_{iu} \quad (22)$$

Models for the quasi-steady force are well studied, (see e.g. [6] for a review), and the main closure needed is for the inviscid unsteady force. F_{iu} can be decomposed into two pieces, a history term and the pressure gradient:

$$F_{iu}(t) = \int_{-\infty}^t K_{sp} \left(\frac{c_\infty}{R} (t - \chi) \right) \cdot \left(\frac{D(m_f u)}{D\chi} - \frac{d(m_f v)}{d\chi} \right) d \left(\frac{c_\infty}{R} \chi \right) + \int_V \rho_f \frac{Du}{Dt} dV \quad (23)$$

For a given post-shock Mach number M_∞ , the force kernel $K_{sp,M_\infty}(\tau)$ can be described. Analytic methods [15] succeeded in the limit $M_\infty \rightarrow 0$,

$$K_{sp,M_\infty \rightarrow 0}(\tau) = \exp(-\tau) \cos(\tau) \quad (24)$$

and [11] set out a numerical technique to determine the force kernel for finite Mach numbers $M_\infty > 0$. It was found the force kernel has a significant dependence on the post-shock Mach number.

In order to calculate Eq. [23], Parmar [4] supposed a zero-width shock (equivalent to the assumption of shock interface to be small compared to the particle size), and performed up/downstream averaging to compute derivative terms in the particle reference frame. Defining $V_2(t)$ to be the volume of the particle subsumed by the shock, $V_1(t)$ its complement so that $V = V_1(t) + V_2(t)$; $A(t)$ to be the cross-sectional area of the particle cut by the shock location; and the shock location w.r.t. the front of the particle $x_s(t)$, allows geometric arguments to determine the dynamic quantities. Namely we can define the fluid velocity and acceleration in the particle reference frame as

$$\frac{u(x_s(t)) - u_1}{u_2 - u_1} = \frac{V_2}{V} \quad (25)$$

$$\frac{1}{u_2 - u_1} \frac{d u(x_s(t))}{dt} = \frac{A(t)}{V} u_s \quad (26)$$

with the fluid density and its derivative defined similarly.

Finally, the pressure gradient is determined by averaging upstream and downstream conditions

$$\int_V \rho_f \frac{Du}{Dt} dV = - \int_V \frac{\partial p}{\partial x} dV \approx (p_2 - p_1) A(t) \quad (27)$$

After these approximations, one only needs $K_{sp,M_\infty}(t)$. In [4], the authors used their previous work to compute

$$K_{sp,M_\infty}(t) = \frac{1}{m_f a} \frac{d}{d\tau} F_{pert} \quad (28)$$

where F_{pert} is a force on the particle after imposing an acceleration a in an otherwise steady-state flow. Instead of computing this additional simulation, we parameterize $K(t)$ as stated in the text.